

Question	Answer	Marks
4(a)	wave passing through gap / aperture	B1
	(wave) spreads (out)	B1
4(b)(i)	$n\lambda = d \sin \theta$	C1
	$\theta = \sin^{-1} [(540 \times 10^{-9}) / (5.0 \times 10^{-6})] = 6.2^\circ$	A1
4(b)(ii)	$\theta = 12^\circ$	A1
4(b)(iii)	peaks / maxima shown at $\theta = 0, \pm 6^\circ$ and $\pm 12^\circ$	B1
	peaks / maxima and zero intensity in between	B1
	central peak / maxima at intensity I_0 and the other peaks all equal to or less than I_0	B1
4(c)	peaks / maxima all at the same angles (as before)	B1
	intensity (of all peaks / maxima) halved	B1

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